1 Matrices

Given the general system of m linear equations in n unknowns

$$\begin{aligned} A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n &= y_1 \\ A_{21}x_1 + A_{22}x_2 + \cdots + A_{2n}x_n &= y_2 \\ \vdots \qquad\quad & \vdots \qquad\quad \vdots \quad \vdots \\ A_{m1}x_1 + A_{m2}x_2 + \cdots + A_{mn}x_n &= y_m, \end{aligned} \tag{1.1}$$

we wish to form linear combinations of these equations in such a way that we are guaranteed to produce an equivalent system which is easier to solve. In this section, we will formalize this process and make precise the kind of system at which we wish to arrive.

In forming linear combinations of the equations in (1.1), notice that we are actually only computing with the coefficients A_{ij} and scalars y_i, with the variables x_j more or less acting as placeholders. We shall therefore abbreviate the system by

$$AX = Y \tag{1.2}$$

where

$$A = \begin{bmatrix} A_{11} \cdots A_{1n} \\ \vdots \qquad \vdots \\ A_{m1} \cdots A_{mn} \end{bmatrix} \tag{1.3}$$

$$X = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \text{ and } Y = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} \tag{1.4}$$

Definition 1.5 (Coefficient matrix)**.** A rectangular array of numbers as in (1.3) is called a *matrix*. In (1.3), the matrix A is called the *coefficient matrix* of the system. The mn numbers A_{ij} (which are the coefficients of the equations in (1.1)) are called the *entries* (or *matrix elements*) of the matrix A. Since A has m rows and n columns, it is said to be an $m \times n$ matrix. Note that the number of *rows* is always listed first. [1]

Example 1.6 (Coefficient matrix of a linear system)**.** The matrix of coefficients for the linear system $\begin{aligned} 2x_1 - x_2 + x_3 &= 0 \\ x_1 + 3x_2 + 4x_3 &= 0 \end{aligned}$ is $\begin{bmatrix} 2 & -1 & 1 \\ 1 & 3 & 4 \end{bmatrix}$.

1. Write the coefficient matrix for the homogeneous linear system $\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_2 - 8x_3 &= 0 \\ 5x_1 - 5x_3 &= 0. \end{aligned}$

[1] For now $AX = Y$ is simply a shorthand notation for the system (1.1). We will soon define a multiplication operation for matrices such that the product of A and X is Y.

2 Elementary row operations

We now consider operations on the rows of the matrix A which correspond to forming linear combinations of the equations in the system $AX = Y$. We only wish to consider operations which lead to an *equivalent* system of equations. We restrict our attention to three *elementary row operations*.

Definition 2.1 (Elementary row operations)**.** The three *elementary row operations* are the following:

1. (Scaling) Multiply all entries in a row by a *nonzero* scalar. $(R_i \mapsto cR_i)$
2. (Replacement) Replace one row by the sum of itself and a scalar multiple of another row. $(R_i \mapsto R_i + cR_j)$
3. (Interchange) Interchange two rows. $(R_i \leftrightarrow R_j)$

More formally, we can view each elementary row operation as a *function* e which takes an $m \times n$ matrix A to an $m \times n$ matrix $e(A)$. The function is specified on the matrix elements A_{ij} of A explicitly in each of the three cases above as follows:

$$1.\ e(A)_{ij} = \begin{cases} A_{ij} \text{ if } i \neq r, \\ cA_{rj} \text{ if } i = r, \end{cases} \qquad 2.\ e(A)_{ij} = \begin{cases} A_{ij} \text{ if } i \neq r, \\ A_{rj} + cA_{sj} \text{ if } i = r, \end{cases} \qquad 3.\ e(A)_{ij} = \begin{cases} A_{ij} \text{ if } i \neq r,s, \\ A_{sj} \text{ if } i = r, \\ A_{rj} \text{ if } i = s. \end{cases}$$

Remark 2.2. Note that the rows of $e(A)$ are linear combinations of the rows of A. We will now prove that each of these three elementary row operations has an inverse which is itself an elementary row operation of the same type. This implies that the rows of A are also linear combinations of the rows of $e(A)$, and therefore the corresponding systems are equivalent (have the same solution set).

Example 2.3. The matrix $\begin{bmatrix} 0 & -2 & 5 \\ 1 & 4 & -7 \\ 3 & -1 & 6 \end{bmatrix}$ is transformed into $\begin{bmatrix} 1 & 4 & -7 \\ 0 & -2 & 5 \\ 3 & -1 & 6 \end{bmatrix}$ by the elementary row operation $e = R_1 \leftrightarrow R_2$. The inverse elementary row operation changing it back is the same operation: $e^{-1} = e$.

2. For each pair of matrices below, find the elementary row operation that transforms the first matrix into the second, and then find the inverse row operation that transforms the second matrix into the first.

   (a) $\begin{bmatrix} 1 & 3 & -4 \\ 0 & -2 & 6 \\ 0 & -5 & 9 \end{bmatrix}, \begin{bmatrix} 1 & 3 & -4 \\ 0 & 1 & -3 \\ 0 & -5 & 9 \end{bmatrix}$

   (b) $\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 5 & -2 & 8 \\ 4 & -1 & 3 & -6 \end{bmatrix}, \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 5 & -2 & 8 \\ 0 & 7 & -1 & -6 \end{bmatrix}$

   (c) $\begin{bmatrix} 1 & 2 & -5 & 0 \\ 0 & 1 & -3 & -2 \\ 0 & -3 & 9 & 5 \end{bmatrix}, \begin{bmatrix} 1 & 2 & -5 & 0 \\ 0 & 1 & -3 & -2 \\ 0 & 0 & 0 & -1 \end{bmatrix}$

Proposition 2.4 (Elementary row operations are invertible)**.** *To each elementary row operation e there corresponds an elementary row operation e^{-1} of the* same *type as e, such that $e^{-1}(e(A)) = e(e^{-1}(A)) = A$ for all A.*

Proof. We prove this for the scaling operation, leaving the others as an exercise. Let $e = R_i \mapsto cR_i$, where c is nonzero. We will prove that the inverse operation is $e^{-1} = R_i \mapsto \frac{1}{c}R_i$. If $j \neq i$, then

$$e^{-1}(e(A_{jk})) = e^{-1}(A_{jk}) = A_{jk},$$
$$e(e^{-1}(A_{jk})) = e(A_{jk}) = A_{jk}.$$

If $j = i$, then

$$e^{-1}(e(A_{ik})) = e^{-1}(cA_{ik}) = \frac{1}{c}(cA_{ik}) = 1 \cdot A_{ik} = A_{ik},$$
$$e(e^{-1}(A_{ik})) = e(\frac{1}{c}A_{ik}) = c\left(\frac{1}{c}A_{ik}\right) = 1 \cdot A_{ik} = A_{ik}.$$

This proves that $e^{-1}(e(A)) = e(e^{-1}(A)) = A$, hence e is invertible with 2-sided inverse e^{-1}. □

3. Complete the proof of Proposition 2.4.

Definition 2.5 (Row-equivalent matrices)**.** Two $m \times n$ matrices are said to be *row-equivalent* if one can be obtained from the other by a finite sequence of elementary row operations.

To prove the next lemma, recall that if f, g are any invertible functions from a set to itself, then $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$. It follows from this that if $f_1, \ldots, f_n$ is any finite sequence of such functions, then $(f_1 \circ f_2 \circ \cdots \circ f_n)^{-1} = f_n^{-1} \circ \cdots \circ f_2^{-1} \circ f_1^{-1}$. Since function composition is associative, there is no need to insert parentheses in this expression.

Lemma 2.6 (Row-equivalence is an equivalence relation)**.** *Row-equivalence is an equivalence relation. That is, if A, B and C are any $m \times n$ matrices, then they satisfy the following properties*

(i) (Relfexivity) A is row-equivalent to itself;

(ii) (Symmetry) If A is row-equivalent to B, then B is row-equivalent to A;

(iii) (Transitivity) If A is row-equivalent to B, and B is row-equivalent to C, then A is row-equivalent to C.

Proof. We prove reflexivity, leaving the proof of symmetry and transitivity as an exercise. Given any matrix A, let $e = R_1 \mapsto 1 \cdot R_1$. Then $A = e(A)$, which shows that A is row-equivalent to itself. □

4. Complete the proof of Lemma 2.6.

Theorem 2.7 (Row-equivalence implies equivalence for homogeneous systems)**.** *If A and B are row-equivalent $m \times n$ matrices, then the linear systems $AX = 0$ and $BX = 0$ are equivalent (have exactly the same solutions).*

Proof. Since we pass from A to B by a finite sequence of elementary row operations

$$A = A_0 \to A_1 \to \cdots \to A_k = B,$$

by transitivity (see Lemma 2.6) it suffices to prove that the systems $A_jX = 0$ and $A_{j+1}X = 0$ have the same solutions, i.e., that one elementary row operation does not disturb the set of solutions. As explained in Remark 2.2, this is indeed the case. Hence, these two systems are equivalent. □

Example 2.8 (Solving a homogeneous system of equations by elementary row operations)**.** We now demonstrate how to use Theorem 2.7 to solve a homogeneous system of liner equations by elementary row operations.

Consider the homogeneous system of linear equations

$$\begin{aligned} 2x_1 - x_2 + 3x_3 + 2x_4 &= 0 \\ x_1 + 4x_2 - x_4 &= 0 \\ 2x_1 + 6x_2 - x_3 + 5x_4 &= 0. \end{aligned} \tag{2.9}$$

We write the coefficient matrix of the system and apply the following sequence of elementary row operations:

$$\begin{bmatrix} 2 & -1 & 3 & 2 \\ 1 & 4 & 0 & -1 \\ 2 & 6 & -1 & 5 \end{bmatrix} \xrightarrow{R_3 \to -R_1 + R_3} \begin{bmatrix} 2 & -1 & 3 & 2 \\ 1 & 4 & 0 & -1 \\ 0 & 7 & -4 & 3 \end{bmatrix} \xrightarrow{R_1 \to \frac{1}{2}R_1}$$

$$\begin{bmatrix} 1 & -1/2 & 3/2 & 1 \\ 1 & 4 & 0 & -1 \\ 0 & 7 & -4 & 3 \end{bmatrix} \xrightarrow{R_2 \to -R_1 + R_2} \begin{bmatrix} 1 & -1/2 & 3/2 & 1 \\ 0 & 9/2 & -3/2 & -2 \\ 0 & 7 & -4 & 3 \end{bmatrix} \xrightarrow{R_2 \to \frac{2}{9}R_2}$$

$$\begin{bmatrix} 1 & -1/2 & 3/2 & 1 \\ 0 & 1 & -1/3 & -4/9 \\ 0 & 7 & -4 & 3 \end{bmatrix} \xrightarrow{R_3 \to -7R_2 + R_3} \begin{bmatrix} 1 & -1/2 & 3/2 & 1 \\ 0 & 1 & -1/3 & -4/9 \\ 0 & 0 & -5/3 & 55/9 \end{bmatrix} \xrightarrow{R_3 \to -\frac{3}{5}R_3}$$

$$\begin{bmatrix} 1 & -1/2 & 3/2 & 1 \\ 0 & 1 & -1/3 & -4/9 \\ 0 & 0 & 1 & -11/3 \end{bmatrix} \xrightarrow{R_2 \to \frac{1}{3}R_3 + R_2} \begin{bmatrix} 1 & -1/2 & 3/2 & 1 \\ 0 & 1 & 0 & -5/3 \\ 0 & 0 & 1 & -11/3 \end{bmatrix} \xrightarrow{R_1 \to -\frac{3}{2}R_3 + R_1}$$

$$\begin{bmatrix} 1 & -1/2 & 0 & 13/2 \\ 0 & 1 & 0 & -5/3 \\ 0 & 0 & 1 & -11/3 \end{bmatrix} \xrightarrow{R_1 \to \frac{1}{2}R_2 + R_1} \begin{bmatrix} 1 & 0 & 0 & 17/3 \\ 0 & 1 & 0 & -5/3 \\ 0 & 0 & 1 & -11/3 \end{bmatrix}$$

The final matrix is the coefficient matrix of the system

$$\begin{aligned} x_1 + \frac{17}{3}x_4 &= 0 \\ x_2 - \frac{5}{3}x_4 &= 0 \\ x_3 - \frac{11}{3}x_4 &= 0 \end{aligned} \tag{2.10}$$

whose solution set is given by taking $x_4 = t$ to have any value, and then using (2.10) to solve for x_1, x_2, and x_3 in terms of t:

$$\{(-\frac{17}{3}t, \frac{5}{3}t, \frac{11}{3}t, t) : t \in F\}. \tag{2.11}$$

Since the coefficient matrices of (2.9) and (2.10) are row-equivalent, by Theorem 2.7 the systems (2.9) and (2.10) are equivalent, and hence (2.11) is also the solution set of the original system (2.9).

5. Find the solution set of the system

$$\begin{aligned} -x_1 + ix_2 &= 0 \\ -ix_1 + 3x_2 &= 0 \\ x_1 + x_2 &= 0 \end{aligned}$$

by writing its coefficient matrix and performing the following sequence of elementary row operations:

(a) $R_1 \leftrightarrow R_3$

(b) $R_3 \mapsto R_1 + R_3$

(c) $R_2 \mapsto R_2 + iR_1$

(d) $R_2 \mapsto \frac{1}{3+2i} R_2$

(e) $R_3 \mapsto R_3 - (2+i)R_2$

(f) $R_1 \mapsto R_1 - 2R_2$

In the previous examples we were obviously not performing row operations at random. Instead, our choice of row operations was motivated by a desire to simplify the coefficient matrix in a manner analogous to 'eliminating unknowns' in the system of linear equations. Roughly speaking, we use the x_1 term in the first equation of the system to eliminate the x_1 terms in the other equations. Then we use the x_2 term in the second equation to eliminate the x_2 terms in the other equations, and so on, until we obtain the simplest possible equivalent system of equations. Next, we will discuss an algorithm for efficiently carrying out this process.